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Differences Between First- and Second-Year Student Teachers' Practice Self-Efficacy: A Cross-Sectional Study

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Abstract

Do teacher education programs improve students' confidence in their field practice teaching skills? Despite a growing interest in how student teachers' practice self-efficacy (PSE) develops, we know little about the impact of the various components of teacher education programs on PSE. The present study examined whether the first year of teacher education, and particularly the field practice in schools which is directed at training and learning teacher practices, is associated with practice self-efficacy using a targeted measure of PSE for student teachers. Using independent sample *t*-tests and one-way analysis of variance with survey data from 338 students, we show that second-year students have higher PSE than first-year students on most PSE dimensions, with the largest differences being on the PSE dimensions of *Planning and preparation*, *Teaching in itself*, and *Evaluation and development*. In contrast, first-year students scored higher on *Adult collaboration PSE*. Further exploratory analyses showed that English majors had lower *Planning and preparation* and *Teaching in itself PSE* than other majors, whereas Mathematics majors had higher *Adult collaboration PSE*. We also conducted item analysis for the purpose of validating the PSE for both first- and second-year students. The findings advance our knowledge of differences in practice self-efficacy over the first year of teacher education.

Keywords: practice self-efficacy; student teachers; first year; field practice; differences; Rasch analysis; validation; Practice Self-Efficacy Questionnaire (PSEQ)

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1. Introduction

Self-efficacy, within Bandura's social cognitive theory, refers to individuals' beliefs in their capabilities to plan and perform actions necessary to attain a certain outcome (Bandura, 1977, 1997). It has been found to be part of motivation processes, thus co-determining how actions are planned and carried out and how persistent the individual is (Bandura, 1997; Tinto, 2017; Wentzel & Miele, 2009). High self-efficacy has also been found to facilitate cognitive processes in relation to both decision making and learning (Maddux, 2002; Scholz et al., 2002; Schunk & DiBenedetto, 2020).

In relation to learning, Bandura (1977) proposed that students with high self-efficacy for accomplishing a specific task would participate more readily, work harder and be more persistent when facing difficulties than students with low self-efficacy. The relationships between self-efficacy and individuals' feelings, cognitive processes, behavior, and experiences are reciprocal, as one can affect the other and vice versa (Bandura, 1997). Thus, the four

sources of self-efficacy—own experiences of mastery or failure, vicarious experiences of mastery or failure, verbal persuasion (i.e., encouragement and feedback), and the affective states associated with learning—not only affect self-efficacy but are also affected by the student's sense of self-efficaciousness. Self-efficacy is malleable and can be affected both positively and negatively by experiences, while self-efficacy also affects how situations, challenges, learning and so on are experienced.

Within the field of education, self-efficacy has been defined in various contexts and often in role or profession-specific forms, e.g., teacher self-efficacy (TSE), which reflects how a teacher's self-efficacy influences occupational decision making, career choice, career development, and career trajectory (Bandura, 1997; Klassen & Chiu, 2011). Another example is the more general form of academic self-efficacy (ASE), reflecting student learning self-efficacy, most often in relation to academic subjects, which has been shown to influence educational outcomes, persistence and intention to drop out or persist (Bager-Elsborg et al., 2019; Khine & Nielsen, 2022; Richardson et al., 2012). The newest bud on the educational self-efficacy tree is the concept of practice self-efficacy (PSE), also reflecting student learning self-efficacy but in relation to the learning of the practical and performative skills and competences necessary in their future professions, e.g., the teaching profession (Nielsen et al., 2024a, 2024b, 2025; Nielsen & Pettersson, 2025).

As the practice self-efficacy concept is still new, only a few studies exist related to teacher education. Nielsen et al. (2024a) found that student teachers and their field practice teachers had qualitatively different perceptions of which specific teacher and teaching-related practices might challenge student teachers during field practice. Both students and field practice teachers mentioned adaptability in teaching, balancing subject knowledge with pedagogical competence, classroom leadership, and teaching independently. However, the study also found that only the student teachers were preoccupied with their teaching performance, while the field practice teachers emphasized the importance of collaborative practices with parents.

In addition, a study has developed and validated an instrument for the measurement of seven dimensions of PSE within three domains: the Practice Self-Efficacy Questionnaire (PSEQ; Nielsen et al., 2024b; Nielsen & Pettersson, 2025). Nielsen et al. (2024b) further found that first-year student teachers who had teaching experience prior to starting teacher education scored significantly higher on all seven PSE dimensions than student teachers without such experience. In the most recent study, also on first-year student teachers, Nielsen et al. (2025) found that when considering all seven PSE dimensions together in a graphical model, only two dimensions were affected directly by prior teaching experience. The *Differentiation PSE* dimension was influenced negatively, whereas the *Teaching in itself PSE* dimension was affected positively by prior teaching—an impact also moderated by a teaching major.

Despite these insights, we still know very little about whether and how teacher education relates to practice self-efficacy. Typically, a professional teacher education consists of both (a) learning theoretical knowledge and (b) field practice placements in schools, where students practice core skills, which might nurture confidence in these. At present, no study has compared how first- and second-year student teachers differ in their level of practice self-efficacy.

The Current Study

The primary aim of the study was to test the theoretically derived hypothesis that second-year student teachers have higher practice self-efficacy than first-year students; that is, higher scores on each of the seven PSE dimensions. Also, we wanted to explore

whether there were differences in the PSE dimension scores for students with different teaching majors.

The hypothesis that second-year student teachers should have higher practice self-efficacy (PSE) than first-year student teachers was mainly based on Bandura's (1997) theory of self-efficacy, in the sense that if the teacher education program provided students with an optimal learning environment and field practice experiences in schools in terms of mastery experiences, positive vicarious experiences, feedback and encouragement, and positive affective experiences in relation to the teacher and teaching-related practices they encounter in the first year of study, this should affect their practice self-efficacy positively. Previous research also factored into the hypothesis as it had previously been found that teaching experience was related to higher PSE no matter the dimension considered (Nielsen et al., 2024b). Thus, a full years' worth of educational experience, including field practice placement in a school providing experience with various teacher and teaching-related practices, should lead to enhanced PSE, if the educational conditions were optimal.

The exploration of the relationship between teaching majors and PSE dimensions follows from previous findings of teaching major moderating the relationship between prior teaching experience and one of the PSE dimensions, when considering all PSE dimensions in a multivariate graphical model (Nielsen et al., 2025). Thus, we aimed to explore this further by testing directly for differences between students with different teaching majors.

The secondary aim was to conduct a validity study of the psychometric properties of the PSEQ for both first- and second-year student teachers, as the PSEQ had previously only been validated for first-year student teachers. In particular, it was necessary to determine whether PSE dimension scores were invariant across the first two years of teacher education, and if not to employ scores adjusted for study-year DIF to ensure unconfounded comparison of first- and second-year students' scores. For this purpose, we will use the same psychometric models as those used in Nielsen et al.'s (2024b) study, as this will facilitate comparison of the two psychometric studies.

2. Methods

2.1. Instrument

We used the Practice Self-Efficacy Questionnaire (PSEQ; Nielsen & Pettersson, 2025), developed and first validated by Nielsen et al. (2024b). The PSEQ consists of 40 items divided into seven subscales, each measuring a dimension of practice PSE. The seven PSE dimensions belong to one of three domains of practice self-efficacy: the teaching-related PSE domain with the dimensions of *Planning and preparation PSE* (7 items), *Teaching in itself PSE* (7 items), *Class management PSE* (6 items), and *Differentiation PSE* (6 items); the relational PSE domain with the dimensions of *Pupils PSE* (4 items) and *Adult collaborators PSE* (parents and co-teachers) (4 items); and the evaluative and developmental PSE domain with the single dimension of *Evaluation and development PSE* (6 items).

In the PSEQ, students are asked "In your upcoming field practice, how confident are you that you have sufficient abilities to…", followed by the item statements, which they rate using a four-point response scale: very confident (3), confident (2), not particularly confident (1), not confident at all (0). The validated Danish items used in the current study are available as a questionnaire (Nielsen & Pettersson, 2025), while both Danish and non-validated English translations are provided in Nielsen et al. (2024b).

A previous validity study with only first-year students (Nielsen et al., 2024b) found that the seven individual subscales of the PSEQ fit either a Rasch model (RM) or a graphical loglinear Rasch model (GLLRM), which accounts for local dependence among items and/or differential item functioning. Only the *Class management* subscale included an item (I4) that functioned differentially, while the *Planning and preparation*, the *Teaching in itself*, the *Adult*

collaborators, and the *Evaluation and development* subscales contained two or three locally dependent items.

2.2. Data Collection and Participants

Data was collected among student teachers at one Danish university college during the two weeks leading up to either their first- or second-year field practice placement. Students completed an online questionnaire that consisted of the PSEQ instrument and a number of demographic and background questions. With the questionnaire, students were informed about (a) the purpose of the study, (b) the voluntary nature of participation, (c) that they could withdraw at any time, and (d) the European data protection regulations. Data was collected from 2022 to 2024 as part of a larger project, and due to the purpose of the current study, data was drawn to obtain a study sample with independent groups of first- and second-year students. Firstly, first-year students who had been included in a previous validity study were excluded. Secondly, first- and second-year students were drawn, so that no student appeared more than once, as we needed a sample with independent groups of students for comparison as well as the item analysis.

Sample size calculations using Gpower 3.1 (d = 0.5, $\alpha = 0.05$, power = 0.95, and one-tailed for comparisons of first- and second-year students, two-tailed for comparisons of students by major) showed that groups should consist of at least 88 cases for the comparison of first- and second-year students, and at least 105 cases for comparisons of students with differing majors.

The Danish teacher education (Act on the Education of Teachers of 2023, 2023) is a four-year-long program leading to a Professional Bachelor's degree. Field practice is included in the program and amounts to 40 ECTS points (European Credit Transfer System), which is equivalent to two-thirds of a full-time study year. Each of the four years of the program involves a field practice placement. During the first and second year of study, two periods of time with one-day visits of field practice are placed prior to and after a three-week full-time field practice placement. The first- and second-year field practice is completed in the same school.

The study sample comprised 338 student teachers, who were in their first or second year of teacher education, with an almost equal distribution across study years (Table 1). The vast majority of participants attended the regular teacher education program. The majority of students (63%) had chosen Danish language as their major teaching subject, whereas 22% had selected Mathematics and 15% English. In the Danish teacher education program, students choose their major among these three subjects, and they receive training in them for two years. The mean age of the students was 24.4 years, and two-thirds were female.

Table 1. Characteristics of the student sample (N = 338).

	Frequency (%)
Teacher education program	
Regular	319 (94.7)
Other [a]	19 (5.3)
Year of study	
1st year	184 (54.4)
2nd year	154 (45.6)
Major teaching subject	
Danish language	213 (63.0)
Mathematics	75 (22.2)
English	50 (14.8)

Table 1. *Cont.*

	Frequency (%)
Gender	
Female	225 (66.6)
Male	111 (32.8)
Other gender identification [b]	2 (0.6)
Mean age (SD), range	24.4 (5.9), 18–68

Notes. [a] includes STEM program, trainee program and distance (online) program. [b] not included in analyses.

The distributions of the seven PSE scores for the study sample are shown in Table 2. With the *Adult collaborators* subscale, the full range of the scale was used, while with the remaining six subscales, the lower end of the scales was not used. The standard deviations (SD) show all subscales provided considerable variation; coefficients of variation (CV; SD/M) ranged between 20% and 39%. The histograms illustrate the distribution of the scores on each PSE dimension.

Table 2. Descriptive statistics for the seven practice self-efficacy dimension subscales.

PSE Subscale (n)	Range	Min	Max	Median (IQR)	Mean	SD	Histogram
Planning and preparation (332)	0–21	3	21	14 (12–17)	14.3	3.71	
Teaching in itself (332)	0–21	6	21	16 (14–18)	15.5	3.12	
Differentiation (332)	0–18	5	18	12 (10–14)	12.1	2.74	
Class management (332)	0–15	4	18	13 (11–15)	12.9	2.67	
Pupils (332)	0–12	4	12	8 (7–10)	8.6	1.91	
Adult collaborators (336)	0–12	0	12	7 (5–9)	7.0	2.72	
Evaluation and development (332)	0–18	4	18	12 (11–15)	12.7	2.95	

Note. IQR = Interquartile range.

2.3. Analyses

First, we conducted item analysis using Rasch measurement models to establish the psychometric properties of the seven practice self-efficacy dimension subscales and to obtain scores adjusted for any differential item functioning related to year of study or teaching major for the subsequent analyses of differences dependent on practice self-efficacy, dependent on these variables (the primary aim of the study).

2.3.1. Item Analysis

The Rasch model (RM; Rasch, 1960) is a parsimonious latent variable model describing the causal relationship between a unidimensional latent variable and item responses. The RM is a member of the Item Response Theory (IRT) models (Fischer & Molenaar, 1995; Christensen et al., 2013). Item responses fitting a Rasch model, whether this contains dichotomous or polytomous items, adhere to the same essential assumptions (Kreiner, 2013; Mesbah & Kreiner, 2013):

- *Unidimensionality*: The items of a scale assess one single underlying latent construct.
- *Conditional independence of items (no local dependence; No LD)*: The response to a single item should be conditionally independent from the response to any other item in the scale given the latent variable.
- *Absence of differential item functioning (no DIF)*: Items and exogenous (i.e., background variables) should be conditionally independent given the latent variable.
- *Overall invariance*: The entire set of items should function invariantly across subgroups.

- *Overall homogeneity*: The rank order of the item parameters should be the same across all persons regardless of their level on the latent variable.

Items in the PSEQ subscales are all ordinal polytomous items; thus, we used the Rasch model for ordered polytomous data. This can be parameterized in different ways. The software used utilizes the power series distribution for estimation but presents various results in different parameterizations. As many researchers are familiar with the Partial Credit Model (PCM; Masters, 1982), we include estimates of the partial credit thresholds and locations in the results.

Sufficiency of the raw summed score distinguishes scales fitting Rasch models from scales fitting other IRT models (Kreiner, 2013). Sufficiency is desirable when a scale is intended for use by practitioners as well as researchers. The Rasch model also facilitates the use of the so-called 'theta scores', which are the person parameter estimates expressed on the logit scale, where the distance between any two values on the scale is equal (often referred to as 'interval-scaled scores' or "equidistant scores"). Depending on the intended use, researchers and practitioners may prefer one over the other, as the summed scores are more easily interpreted, while the theta scores can improve precision. However, the distribution of the scores should also be taken into account, as the theta scores are known to be biased at the extremes, and thus a distribution with many cases at one or both extremes will be detrimental to precision, in which case the summed scores should be preferred (Kreiner & Christensen, 2013).

If fit to the RM is rejected, it is possible to achieve close to optimal measurement (i.e., essential validity) if the only departures from the RM are uniform DIF and/or uniform LD (Kreiner & Christensen, 2007), as such departures can be adjusted for within the extended GLLRM. If a GLLRM only included uniform LD among some items, the sufficiency of the sum score is not affected, but the reliability of the scale will be affected negatively to some degree (Marais, 2013). If a GLLRM includes items with uniform DIF, the sum score can still be a sufficient statistic for the latent score if the DIF is resolved through DIF-equating, thus eliminating the confounding effect of the DIF-variables (Kreiner, 2007).

For more details on Rasch and graphical loglinear Rasch models, we refer the reader to the initial validity study of the PSEQ and Nielsen & O'Neill's study of a generalized critical thinking scale for easy-to-read explanations (Nielsen et al., 2024b; Nielsen & O'Neill, 2025), as well as the methodological literature referenced in this section and the following section with test statistics for more technical and statistical explanations.

2.3.2. Strategy of Analyses and Test Statistics Used

The RM is characterized by the sufficiency of the raw score for the person parameter estimates—a property exclusive of Rasch models. Because of this, the estimation of parameters and inference concerning the fit of items to the RM can be conditional. Conditional inference eliminates person parameters by using the conditional distributions of items that do not depend on the person parameters, and therefore conditional inference does not need assumptions concerning the distribution or sampling of persons or the person parameters (Christensen, 2013; Kreiner, 2013, 2025; Rasch, 1961). Item parameters were calculated using conditional maximum likelihood (CML) estimates.

The DIGRAM software (Kreiner, 2003; Kreiner & Nielsen, 2013, 2023) was used for all item analyses. As all inference in DIGRAM is conditional the person parameters are not used when testing the model, but only for analysis of reliability and targeting, used once fit to the model has been tested in the following ways (c.f. the below).

The same strategy of analysis was used for each of the seven subscales. All tests used tested the null hypothesis that item responses fit the Rasch model against different alternatives. First, the full set of original items were tested against the requirements of the

RM using the following test statistics: conditional likelihood ratio (CLR) tests of overall homogeneity and of invariance were used, respectively, to compare item parameters across low and high scoring groups and to compare the full set of item parameters in subgroups defined by exogeneous variables, one exogenous variable at a time (Andersen, 1973). Item fit statistics comparing the observed and expected correlations under the model between single items and the restscore (i.e., the total score without the item in question) were used to assess individual item fit (Christensen, 2013; Kreiner, 2011). Kelderman's (1984) likelihood ratio test was used to test the local independence of items and absence of DIF, and if evidence of LD or DIF was discovered, the magnitude of these was informed by partial Goodman–Kruskal gamma coefficients (Goodman & Kruskal, 1954).

Second, if evidence of uniform local dependence and/or uniform DIF (Hanson, 1998) was found in a subscale, we included interaction terms for these in a GLLRM and proceeded to test whether there was fit to this model using the same fit statistics as for the RM (c.f. the above).

Since the relevant independent variables for comparing PSE scores were year of study and major (see Section 2.3.2) and the seven PSE subscales had not previously been tested for DIF or invariance in relation to these variables, these variables were included in the item analysis as exogenous variables. In this way, we would be able to equate the scale scores and ensure that the subsequent comparisons of scale scores for subgroups were not confounded by DIF related to study year or major, if DIF was discovered (Kreiner & Christensen, 2007; Kreiner, 2007; Kreiner & Nielsen, 2023).

The Benjamini–Hochberg procedure was used to adjust for false discovery rate (FDR) due to multiple testing (Benjamini & Hochberg, 1995), while the strength of the evidence against the respective measurement models was evaluated on a continuum distinguishing between weak ($p < 0.05$), moderate ($p < 0.01$) and strong ($p < 0.001$) evidence, as recommended by Cox et al. (1977).

After achieving fit of the item responses to either a Rasch or a graphical loglinear Rasch model, person parameters were estimated and reliability and targeting were assessed. Person parameters were estimated using weighted maximum likelihood estimation, as these are less biased than maximum likelihood estimates (Kreiner & Christensen, 2013; Warm, 1989). For analyses of reliability and targeting, we assumed the distribution of the person parameters (measures of theta) to be normal, estimated the mean and standard deviation of this distribution, and used this distribution to assess reliability and targeting. To estimate the reliability, we used Hamon and Mesbah's Monte Carlo method for reliability in Rasch models (Hamon & Mesbah, 2002) generating simulated person parameter data for 20,000 persons. Using the simulated data, we then estimated reliability as the ratio of the variances of the expected and observed scores, and the Person Separation Index (PSI) as the measure of reliability for the theta scale (Kreiner & Christensen, 2013).

Reliability was calculated using Hamon and Mesbah's (Hamon & Mesbah, 2002) Monte Carlo method, and by way of the interaction terms for ant LD in loglinear Rasch models, we could also take this into account and adjust reliability accordingly. Targeting of both summed scores and theta scores was assessed by comparison of the mean scores to the target values (target is the location on the scale, where most information is available/where the standard error of measurement is the lowest). Targeting of theta was further assessed by the test information target index (the mean test information divided by the maximum test information for theta) and the root mean squared error (RMSE) target index (the minimum standard error of measurement (SEM) divided by the mean SEM for theta) (Kreiner & Christensen, 2013). The target and SEM of the observed score were also calculated.

2.3.3. Differences in Practice Self-Efficacy Scale Scores

Differences in mean practice self-efficacy scale scores were assessed for subgroups defined by study year (the primary independent variable) and by major (the secondary independent variable). For this purpose, we used the summed scores and ran simple Student's *t*-tests for independent groups with 1000 bootstrap samples. For scales with DIF, the scores were adjusted to take the DIF into account before testing for differences in scores. Approximate normality of the PSE scores in the sample, as an expression of normality in the population, was evaluated using P-P plots. We found no indications against using *t*-tests on the sum scores (c.f. the histograms included in Table 2). However, as major had three subgroups, we first conducted a simple analysis of pairwise collapsibility to test whether subgroups all differed, if any two subgroups did not differ and thus should be collapsed and compared to the third subgroup, or if all subgroups did not differ and thus should be collapsed to a single group. The Benjamini and Hochberg (1995) procedure was used to set the critical *p*-value for collapsing of subgroups, and thus the comparison with the highest *p*-value above the critical level was collapsed in a stepwise procedure (Kreiner & Nielsen, 2023). We used one-sided *p*-values for testing the directional hypotheses related to study year, while we used two-sided *p*-values for the exploratory tests of differences by major.

In addition to the mean differences in scale scores and *p*-values, we calculated 95% confidence intervals for each subgroup mean to ascertaining whether 95% CI's for groups compared overlapped; they should not if there is a true difference in subgroup means. We also calculated standardized effect sizes for the differences using the pooled variance (i.e., Cohen's *d*). Cohen's *d* was interpreted as follows: 0.2 = small effect, 0.5 = medium effect, 0.8 = large effect (Cohen, 1988). *p*-values were interpreted as follows: $p > 0.05$ = no evidence of differences, $p < 0.05$ = weak evidence of difference, $p < 0.01$ = medium strong evidence of difference, and $p < 0.001$ = strong evidence of difference.

3. Results

3.1. Item Analysis by Rasch and Graphical Loglinear Rasch Models

The item analysis by Rasch and graphical loglinear Rasch models revealed that the seven PSEQ subscales for the current sample of first- and second-year student teachers performed psychometrically on par with the initial validation study (Nielsen et al., 2024b). The main exception was the discovery of a few additional locally dependent items in two subscales, and DIF related to study year in three subscales. The latter had not been investigated by Nielsen et al. (2024b) as they only included first-year students.

Tables 3 and 4 show the fit of each of the seven PSE dimension scales to either the pure Rasch model or a graphical loglinear Rasch model which included and adjusted for locally dependent items and/or items functioning differentially relative to study year. There was no evidence against global homogeneity (i.e., the set of items in each scale functioned the same for low and high scorers), nor any real evidence against invariance (i.e., the set of items functioned the same across subgroups defined by study year and majors) in the final models, once local dependence and DIF had been accounted for. The only evidence against invariance was found in the *Planning and preparation PSE* subscale, but the evidence was weak, and thus only warrants attention in future studies (Table 3).

Table 3. Global tests of homogeneity and invariance for the four teaching-related PSE subscales of the PSEQ under the final RMs or GLLRMs.

Tests of Fit	Plan-7 (GLLRM) [a]			Teach-7 (GLLRM) [b]			Cman-6 (RM)			Diff-6 (GLLRM) [c]		
	CLR	*df*	*p*	*CLR*	*df*	*p*	*CLR*	*df*	*p*	*CLR*	*df*	*p*
Global homogeneity	17.2.	32	0.985	35.6	29	0.187	15.2	16	0.509	43.6	31	0.066
Invariance relative to:												
Year	51.7	27	0.003 [+]	21.7	17	0.196	27.1	16	0.040 [+]	28.1	21	0.138
Major	96.8	64	0.005 [+]	67.2	58	0.190	40.5	32	0.145	83.4	62	0.036 [+]

Notes. RM: Rasch model; GLLRM: Graphical loglinear Rasch model; CLR: Conditional likelihood ratio; df: degrees of freedom; *p*: *p*-value. Global homogeneity test compares item parameters in approximately equal-sized groups of high and low-scoring students. The critical limits for the *p*-values after adjusting for false discovery rate were: [+] 5% limit $p = 0.0167$, 1% limit $p = 0.0033$. [a] There is local dependence between items 40 and 41, and between items 41 and 42 as well as DIF relative to study year for item 41. [b] There is local dependence between items 25 and 30 as well as DIF relative to study year for items 25 and 29. [c] There is local dependence between items 1 and 2, and items 17 and 19, as well as DIF relative to study year for items 18 and 19.

Table 4. Global tests of homogeneity and invariance for the two relational PSE subscales and the evaluative and developmental PSE subscales of the PSEQ under the final RM or GLLRMs.

Tests of Fit	Adults-4 (GLLRM) [a]			Pupils-4 (RM)			Eval-6 (GLLRM) [b]		
	CLR	*df*	*p*	*CLR*	*df*	*p*	*CLR*	*df*	*p*
Global homogeneity	18.2	25	0.832	6.6	9	0.681	20.4	33	0.957
Invariance relative to:									
Year	33.2	25	0.125	17.9	9	0.036 [+]	14.1	33	0.998
Major	53.0	50	0.358	25.1	18	0.122	36.8	66	0.999

Notes. RM: Rasch model; GLLRM: Graphical loglinear Rasch model; CLR: Conditional likelihood ratio; df: degrees of freedom; *p*: *p*-value. Global homogeneity test compares item parameters in approximately equal-sized groups of high and low-scoring students. The critical limits for the *p*-values after adjusting for false discovery rate were: [+] 5% limit $p = 0.0167$. [a] There is local dependence between items 10 and 11, and between items 11 and 12. [b] There is local dependence between items 33 and 38, items 34 and 35, and items 36 and 37.

The fit of the individual items to their respective scale models is shown in Supplement S1. Only a single item in the *Planning and preparation PSE* subscale (item 41) had weak evidence against fit to the final model ($p < 0.05$, Table S1a, Supplement S1).

The results of testing the necessity of including interaction terms for locally dependent items and/or items functioning differentially, as well as results of testing for the possible presence of additional locally dependent items and items functioning differentially, are shown in Supplement S2. For the five PSE dimension scales that fitted graphical loglinear Rasch models, which accounted for either just locally dependent items or both locally dependent items and items functioning differentially, there was strong evidence for the necessity of including the respective interaction terms. For all seven PSE dimension scales there was no evidence of any additional local dependence or DIF being required to obtain fit to the respective final models.

Information on item category threshold, locations, midpoints and item targets is included for the interested readers in Supplement S3.

3.1.1. Effect of Differential Item Functioning

Three of the seven PSE subscales had one or two items that functioned differentially relative to study year, while no DIF relative to major teaching subject was discovered (c.f. Supplement S2). In the *Planning and preparation PSE* subscale, second-year students systematically felt more confident in regard to planning short teaching modules (i41) than first-year students did, no matter their level of planning-and-preparation-related practice self-efficacy. In the *Teaching in itself PSE* subscale, it was also the second-year students who systematically felt more confident managing on their own without the field practice teacher (i25) and doing what they had planned during teaching (i29) than first-year students, no

matter their level of teaching-related practice self-efficacy. Finally, in the *Differentiation PSE* subscale, second-year students systematically felt more confident in relation to getting the pupils to take part academically in class (i18) and differentiate the teaching and not merely teach in the way that they were taught (i19) than first-year students, no matter their level of teaching-related practice self-efficacy.

Supplement S4 contains information necessary for conversion between sum scores and theta scores as well as information for the DIF-equation of the sum scores for the three scales with evidence of differential item functioning, i.e., the *Planning and preparation PSE,* the *Teaching in itself PSE,* and the *Differentiation PSE* subscales. To facilitate comparison of subscale scores from first- and second-year student teachers, while taking into account the differential item functioning, an SPSS-syntax that will convert sum scores to theta scores and DIF-equated sum scores is publicly available at Zenodo.org https://doi.org/10.5281/zenodo.19064038 (accessed on 8 April 2026).

3.1.2. Targeting and Reliability

Targeting of the seven PSE subscales was generally very good from a test information perspective, as on average between 72% and 95% of the maximum information was obtained, with only two exceptions; the *Teaching in itself* subscale for second-year student teachers and the *Evaluation and development* subscale for all student teachers where 59% and 57% percent of the maximum information was obtained (Table 5). Reliability of the summed scores was satisfactory for statistical analysis, i.e., above 0.70, except for the *Adult collaborators* subscale, where it was only 0.66. The same was true for the reliability (PSI) of the theta scale.

3.2. Differences in PSE Scores for First- and Second-Year Student Teachers

For five of the seven PSE dimensions, second-year students, as expected, had higher scores than first-year students. In contrast, first-year students scored higher than second-year students on the last two PSE dimensions. However, not all differences were significant and effect sizes varied (Table 6). Thus, our hypothesis that second-year students would have higher PSE dimension scores than first-year students was not entirely confirmed.

Large (more than 1 scale point) and highly significant differences in the hypothesized direction (i.e., second-year students scored higher than first-year students) were found for *Planning and preparation PSE*, *Teaching in itself PSE*, and *Evaluation and development PSE*, while differences were smaller (just under 1 scale point) though significant for *Class management PSE*. The effect sizes, Cohen's *d*, ranged from small (i.e., 0.318 for *Class management PSE*), to close to moderate (i.e., 0.477, 0.437 and 0.441 for *Planning and preparation PSE*, *Teaching in itself PSE* and *Evaluation and development PSE*, respectively).

For the seventh PSE dimension, *Adult collaborators PSE*, a large and highly significant difference was also found, but in the opposite direction than expected. Thus, first-year students had higher scores on the *Adult collaborators PSE* dimension than did second-year students. This difference was the largest among the seven dimensions (d = −0.687).

For *Differentiation PSE* and *Pupils PSE* scales, differences were insignificant and the estimated effect sizes were either very small or non-existent. Thus, students did not appear to differ on these dimensions by study year.

Table 5. Targeting and reliability of the seven subscales of the PSEQ (N = 336).

	Theta (*θ*)								Score				
Subscale [a]	**Target**	**Mean**	**TI Mean**	**TI Max**	**TI Target Index**	**RMSE Mean**	**RMSE Min**	**RMSE Target Index**	**PSI**	**Target**	**Mean** [b]	**Target SEM**	**Mean SEM**	***r***
Plan														
1st year	−1.30	0.72	2.818	3.927	0.718	0.608	0.505	0.830	0.81	7.00	13.36	1.98	1.67	0.81
2nd year	−0.34	1.33	2.292	3.211	0.776	0.657	0.558	0.850	0.75	10.87	15.11	1.79	1.56	0.76
Teach														
1st year	−0.86	1.34	1.883	2.242	0.840	0.745	0.668	0.897	0.84	10.24	14.78	1.50	1.36	0.83
2nd year	−1.47	1.88	1.675	2.857	0.586	0.792	0.592	0.747	0.74	9.47	15.85	1.69	1.28	0.74
Cman	−0.37	2.21	1.475	1.794	0.823	0.815	0.747	0.916	0.83	8.89	12.91	1.34	1.21	0.80
Diff														
1st year	−0.34	1.09	1.916	2.134	0.898	0.723	0.685	0.947	0.75	9.22	12.06	1.46	1.38	0.75
2nd year	−0.70	0.73	1.934	2.088	0.926	0.718	0.692	0.964	0.74	8.13	12.42	1.44	1.39	0.73
Adult	−0.71	0.20	2.572	3.226	0.797	0.618	0.557	0.902	0.66	4.48	6.99	1.80	1.60	0.66
Pupils	−2.29	1.47	0.997	1.119	0.891	0.979	0.945	0.965	0.77	4.58	8.58	1.06	0.99	0.72
Eval	−1.53	1.34	1.906	3.342	0.570	0.720	0.547	0.760	0.78	6.17	12.77	1.83	1.37	0.79

Notes. *θ*, theta; Targeting and reliability is provided for groups defined by differential item functioning (DIF) variables and subgroups in variables associated with the latent PTSD variable; Tl, test information; RMSE, root mean squared error of the estimated theta score; SEM, standard error of measurement of the observed score; *r*, reliability; Plan, Planning and preparation; Teach, Teaching in itself; Cman, Class management; Exp, teaching experience; Diff, Differentiation; Adult, Adult collaborators; Eval, Evaluation and development. [a] In subscales with differential item function, results are shown for the relevant subgroups. [b] based on DIF-equated scores.

Table 6. Comparison of PSE subscale scores for 1st- and 2nd-year student teachers by independent sample *t*-tests.

			95% CI					
Subscales and Year Groups	**Mean**	**SD**	**Lower**	**Upper**	***t***	***df***	***p*** [b]	**Cohen's *d***
Planning and preparation [a]								
1st-year students	13.34	3.78	12.84	13.90	4.324	330	<0.001	0.477
2nd-year students	15.06	3.39	14.51	15.63				
Teaching in itself [a]								
1st-year students	14.77	3.28	14.31	15.26	3.053	330	0.001	0.437
2nd-year students	15.82	2.88	15.36	16.29				
Class management								
1st-year students	12.51	2.66	12.11	12.87	2.887	330	0.002	0.318
2nd-year students	13.35	2.63	12.89	13.76				
Differentiation [a]								
1st-year students	12.05	2.79	11.63	12.45	1.603	330	0.055	0.177
2nd-year students	11.55	2.85	11.11	12.06				
Adult collaborators								
1st-year students	7.76	2.62	7.37	8.17	−6.227	334	<0.001	−0.682
2nd-year students	6.00	2.53	5.59	6.43				
Pupils (relational)								
1st-year students	8.55	1.95	8.28	8.84	0.095	330	0.462	0.010
2nd-year students	8.57	1.87	8.26	8.88				
Evaluation and development								
1st-year students	12.18	3.04	11.45	12.61	3.997	330	<0.001	0.441
2nd-year students	13.45	2.69	13.00	13.87				

Notes. There was no evidence against homogeneity of variance across subgroups for any of the scales, and thus the standard *t*-test was used. [a] For these subscales, scores adjusted for DIF were used. [b] *p* is one-sided.

3.3. Differences in PSE Scores for Student Teachers with Different Majors

We did not have any a priori expectations on differences in PSE dimension across the students' three teaching majors; rather, we wanted to explore this. The initial collapsibility analyses (see Supplement S5 for details) showed no significant differences for students with different majors on four of the PSE dimensions, *Class management PSE*, *Differentiation PSE*, *Pupils PSE*, and *Evaluation and development PSE*. Thus, a common (i.e., collapsed) mean value could be calculated for each of them (Table 7).

Table 7. Comparison of PSE subscale scores for student teachers with different majors by pairwise collapsibility analysis and independent sample *t*-tests.

Subscales and Majors	**Before Collapsing** [a]		**After Collapsing** [a]		95% CI					
	Mean	**SD**	**Mean**	**SD**	**Lower**	**Upper**	***t***	***df***	***p*** [c]	**Cohen's *d***
Planning and preparation [b]										
Mathematics	14.48	3.79								
Danish	14.39	3.48	14.42	3.56	13.99	14.86	3.691	330	<0.001	0.571
English	12.34	4.08	12.34	4.08	11.21	13.35				
Teaching in itself [b]										
Mathematics	15.74	3.32								
Danish	15.43	2.82	15.51	2.96	15.19	15.85	3.824	330	<0.001	0.592
English	13.68	3.73	13.68	3.73	12.63	14.72				
Class management										
Mathematics	13.20	2.66								
Danish	12.97	2.56	12.89	2.67	-	-	-	-	-	-
English	12.04	3.05								
Differentiation [b]										
Mathematics	12.17	2.89								
Danish	11.79	2.73	11.83	2.82	-	-	-	-	-	-
English	11.47	3.08								

Table 7. *Cont.*

Subscales and Majors	Before Collapsing [a]		After Collapsing [a]		95% CI					
	Mean	SD	Mean	SD	Lower	Upper	*t*	*df*	*p* [c]	Cohen's *d*
Adult collaborators										
Mathematics	7.79	2.86	7.79	2.86	7.13	8.48				
Danish	6.74	2.69	6.73	2.64	6.41	7.05	3.004	334	<0.001	0.394
English	6.69	2.42								
Pupils (relational)										
Mathematics	8.73	2.00								
Danish	8.63	1.85	8.56	1.91	-	-	-	-	-	-
English	8.02	1.98								
Evaluation and development										
Mathematics	12.89	3.20								
Danish	12.88	2.77	12.75	2.95	-	-	-	-	-	-
English	11.98	3.24								

Notes. There was no evidence against homogeneity of variance across subgroups for any of the scales, and thus the standard *t*-test was used. [a] Test statistics from the collapsibility analysis are shown in Supplement S5. [b] For these subscales, scores adjusted for DIF were used. [c] *p* is two-sided.

For the remaining PSE dimensions, two majors' subgroups were collapsed to form a single group with a common mean score: students who had chosen Mathematics and Danish as their major did not differ in their *Planning and preparation PSE* or their *Teaching in itself PSE*, and were subsequently compared with the students with English as their major. With the *Adult collaborators PSE* dimension, it was students with Danish or English as their major who did not differ and thus were collapsed into a single group and subsequently compared to the group of students with Mathematics as their major. After collapsing, *t*-tests revealed large differences in these three PSE dimensions. Thus, for the *Planning and preparation PSE* and the *Teaching in itself PSE* dimensions, students with English as their major scored approximately two points lower than students who had chosen Danish or Mathematics as their major, and the effect sizes were moderate ($d = 0.571$ and $d = 0.592$, respecitvely). For the *Adult collaborators PSE* dimension, students who had chosen Mathematics as their major scored approximately one scale point higher than the students who had chosen Danish or English as their major, while the effect size was small to moderate ($d = 0.394$).

4. Discussion

The primary and theoretically-derived hypothesis we wanted to test was whether second-year student teachers had higher practice self-efficacy than first-year students, as would be expected from previous research and Bandura's (1997) theory of self-efficacy. That is, expected, if the teacher education program provides students with an optimal learning environment and sufficient field practice experiences in schools in terms of mastery experiences, positive vicarious experiences, feedback and encouragement, and positive affective experiences in relation to the teacher and teaching-related practices they encounter. Secondarily, based on previous research, we wanted to explore whether there were differences in the PSE dimensions scores for students with different teaching majors. Finally, as the PSEQ had not previously been validated for both first- and second-year student teachers, and we did not know whether PSE dimension scores were invariant across the first two years of teacher education. Therefore, we first conducted item analyses using Rasch and graphical loglinear Rasch models to assess, and possibly adjust for, any DIF.

4.1. The Psychometric Results

We briefly discuss the psychometric results before turning to the main results.

The item analysis showed that the psychometric properties of three of the seven PSE dimension subscales were comparable to what has previously been found for only first-year student teachers (Nielsen et al., 2024b). Thus, the *Pupils PSE* and the *Adult collaborators*

PSE subscales respectively fitted an RM and a GLLRM with the same local dependence between items in the latter model, as was the case for first-year students in Nielsen et al. (2024b). The *Class management PSE* subscale fitted an RM in this study, whereas it fitted a GLLRM with DIF relative to teaching experience prior to teacher education for first-year students (Nielsen et al., 2024b). As we did not include prior teaching experience, this was no surprise.

For the remaining four PSE dimension subscales, the results of the item analysis varied from what had previously been found for first-year student teachers by Nielsen et al. (2024b), primarily due to items in three subscales functioning differentially relative to study year, but also due to additional local dependence among some items in two subscales. Additional local dependence was found among two pairs of items in the *Differentiation PSE* subscale, which had previously been reported to fit a pure RM for first-year students and among three item pairs in the *Evaluation and development PSE* subscale, which had previously been reported to include local dependence among another pair of items (Nielsen et al., 2024b).

Differential item functioning relative to study year was found for three of the seven PSE dimension subscales: *Planning and preparation PSE*, the *Teaching in itself PSE*, and the *Differentiation PSE* subscales. In all three cases, the DIF was such that second-year student teachers systematically indicated greater confidence in the practices described in the items, independent of their level of the respective PSE dimension, than did first-year students. These DIF results intuitively make sense as the items involved reflective practice content such as planning short teaching modules (I41 in the *Planning and preparation PSE* subscale), to "manage on my own" without the field practice teacher and do what I had planned during teaching (I25 and I29 in the *Teaching in itself PSE* subscale), and getting the pupils to take part academically in class as well as differentiating teaching and not merely teach in the way that they were taught (I18 and I19 in the *Differentiation PSE* subscale).

Nielsen et al. (2024b) did not report any differential item functioning or lack of invariance related to the time of measurement, which was shorter than in the current study. Thus, systematic differences in the experience or perception of specific-teacher- and teaching-related practices among student teachers are not present across the shorter time span difference from before field practice to after a time period of approximately three months with single day visits during the first year of study in Nielsen et al.'s (2024b) study, but only show up over the longer time span difference of one year from before the field practice in the first year of study. Future research should focus on exploring this further through cross-sectional studies including third- and fourth-year students to uncover whether shifts in perception or experience of more or other teacher and teaching-related practices occur across the teacher education program. Such studies could successfully be illuminated by having students in different study years participate in focus groups to discuss their understanding of and experience with the specific teacher or teaching-related activities where such systematic differences are found.

Reliability of the summed scores as well as the theta scale (i.e., the PSI) were comparable to what has previously been reported for first-year student teachers (Nielsen et al., 2024b), and for the most part satisfactory for statistical analysis. Thus, only the *Adult collaborators PSE* subscale had reliability below 0.70 (i.e., 0.66). This would have been considered an issue if the study were a longitudinal study examining change, but as the current study examines differences between independent groups, it is of less consequence. Targeting of five of the seven PSE dimension subscales was very good, while it was less than satisfactory for the *Teaching in itself* subscale for second-year student teachers and the *Evaluation and development* subscale for all student teachers. This result is new, as the previous validation study for first-year student teachers showed very good targeting for all

subscales, except the *Planning and preparation PSE* subscale, where targeting was acceptable but no more than that.

4.2. Differences in PSE Scores for First- and Second-Year Student Teachers

With regard to the main hypothesis that second-year students would have higher PSE dimension scores than first-year students, this was only partially confirmed, as the predicted differences were only found for four of seven PSE dimension subscales.

Second-year students scored significantly higher than first-year students on *Planning and preparation PSE*, *Teaching in itself PSE*, *Class management PSE*, and *Evaluation and development PSE*. The differences varied from approximately 0.75 points to 1.5 points (*Planning and preparation PSE*), and effect sizes were small to medium.

The largest differences (more than one point) between first- and second-year students in the hypothesized direction were on *Planning and preparation PSE*, *Teaching in itself PSE*, and *Evaluation and development PSE*. These are all rather different dimensions of teacher and teaching-related practices, but also in some ways similar. These similarities may be part of the mechanisms behind the differences. In the teacher education program, a course on pedagogics and didactics is placed in the first year of study and the course and its teachers are involved in both the preparation for field practice placements as well as the post-processing of the students' experiences. The course provides theoretical foundations for planning and preparing teaching, as well as evaluation and development of teaching competencies. Thus, the students are provided with ample opportunities for gaining experience with and feedback on such practices throughout the study year, not only in the field practice placements. As students are placed in schools in groups of three to five students, they also gain vicarious experience with planning and preparation, as well as evaluation and development, through their fellow students' practices. Thus, although students do not always have the opportunity to evaluate their teaching with the pupils or the pupils' learning gains, they gain some experience with formative evaluation as they participate as observers—even if the students do not always perceive this as an experience with evaluation, which can support the development of their teaching. In the field practice placements, all students gain their own experience with teaching and receive feedback on their teaching; they have vicarious experiences of teaching, and all of these activities are associated with affective states. The difference in *Teaching in itself PSE*, with the second-year students scoring the highest, thus suggests that these experiences predominantly are positive. This is supported by previous research that reported a direct effect of teaching experience prior to starting teacher education on *Teaching in itself PSE* prior to field practice for first-year students (Nielsen et al., 2025). The findings are in line with the objectives for the first-year field practice:

> "*The student can:*
>
> - *identify and collect relevant knowledge on teachers' planning and teaching,*
> - *plan, execute and evaluate short sequences of teaching, and is able to account for their own didactical considerations,*
> - *understand and reflect on their own role as a teacher.*
>
> *Furthermore, the student is expected to gain initial experience with investigation and description of their own practices and the practices of others, with observation, and with investigations of issues and challenges related to being a teacher.*" (National curriculum, 2025, p. 6, first author's translation)

The smallest difference between first- and second-year students was in *Class management PSE*. Compared to the other PSE dimension, where second-year students scored higher than first-year students, this result may be found in the organization of the field

practice placements. The student teachers are placed in schools in groups of three to five students, and they are rarely responsible for leading or managing a class entirely on their own. This organization makes it a challenge for the student teachers to gain independent experience in this. Thus, a strengthening or weakening of their *Class management PSE* will rely on vicarious experiences and affective states in relation to these vicarious experiences (Bandura, 1997; Bandura et al., 1963; Goodyear, 2014; Hill & Grossman, 2013).

Our hypothesis of second-year students scoring higher than first-year students on all PSE dimensions was clearly rejected for the *Adult collaborators PSE* dimension. On this PSE dimension the reverse was the case, as first-year students scored significantly higher than second-year students. In addition, this was the PSE dimension with the largest difference (1.75 points; d = −0.682). This "reverse" difference in self-efficacy is likely due to a lack of opportunities to gain experiences with collaborative practices involving parents and in-service teachers. From the students' evaluations of the first-year field practice placements at the university college, where we conducted the study, we know that practices involving collaboration with parents are those which have least often been reported as experienced by the student teachers over the last three years (source: internal data). Legislation states that students should gain experience with collaboration with parents; however, the nature, frequency, or time-wise placement in the program of such experiences are not clarified (Act on the Education of Teachers of 2023, 2023). In an ongoing qualitative study specifically on self-efficacy for school-home collaboration, field practice teachers and school headmasters alike have relayed to us that it is difficult to secure all student teachers gain (mastery) experiences in collaborative practices with parents, due to the time-wise organization and the limited duration of the placements (work in progress).

Our hypothesis was not confirmed for the *Differentiation PSE* and *Pupils PSE* dimensions, as no differences were found between first- and second-year students on these. Differentiation of teaching is didactically and pedagogically complex, not only for student teachers but also for trained in-service teachers, and it is closely linked to the composition of the class, i.e., the pupils' different and multifaceted prerequisites and personalities as well as the socio-emotional learning environment of the class (Eikeland & Ohna, 2022). The complexity of differentiation and the knowledge of the pupils necessary to be able to build relationships with the pupils and engage in differentiation of teaching is the most likely explanation for the lack of differences in first- and second-year students' self-efficacy for such practices. The student teachers' time with the pupils in the field practice of the first year of study is limited to 30 days in total and groups of three–five students are placed together in a school. This, suggests that the individual second-year student teachers only have emerging experiences of forming relationships with pupils and engaging in differentiation-related practices—maybe not enough experience to cause a change in their self-efficacy for practices related to forming relationships with the pupils and differentiation of teaching, which could be detected as a difference between the independent groups of first- and second-year students in the current study. This explanation is supported by previous research by Nielsen et al. (2025), who found that teaching experience prior to teacher education had a negative effect on *Differentiation PSE* for first-year student teachers compared to student teachers without such experience. A finding they suggested was due to the prior teaching experiences in schools, which in accordance with self-efficacy theory (Bandura, 1997) serving as a source of adjustment of their self-efficacy in relation to differentiation of teaching.

4.3. Differences in PSE Scores for Student Teachers with Different Majors

With regard to differences in PSE dimension scores across the three possible teaching majors, we expected based on previous findings (Nielsen et al., 2025) that there would be

differences in the *Teaching in itself PSE* subscale, while we had no specific expectation as to how these would present themselves, but we had no specific expectations for the other six PSE dimension subscales.

We found that student teachers who had chosen English as their major teaching subject scored significantly *lower* on both the *Planning and preparation PSE* subscale and the *Teaching in itself PSE* subscale than the combined group of students, who had chosen Danish or Mathematics as their teaching major. The differences were in both cases close to 2 points on the respective subscales, with medium effect sizes in both cases. A possible explanation for these results may be that English majors are planning and preparing as well as practicing to teach partly in a second language. This adds complexity to these activities and may be a source of lowered self-efficacy for the included teacher and teaching-related practices in their field practice in schools. This explanation is supported—at least for *Teaching in itself PSE*—by previous research by Nielsen et al. (2025), who found that while teaching experience prior to teacher education had a positive effect on *Teaching in itself PSE* for first-year student teachers, this effect was much larger for English majors than majors in Danish and Mathematics.

We further found that student teachers who had chosen Mathematics as their major teaching subject scored significantly *higher*—by approximately 1 point and a small effect size—on the *Adult collaborators PSE* subscale than the combined group of students, who had chosen Danish or English as their teaching major. We have not located research which could enlighten this finding, nor do we have a practice-based explanation. This is thus a finding which could successfully be explored more qualitatively in a future study.

Two other previous studies into student teacher's teacher self-efficacy did *not* vary by major teaching subject, but rather across vocational and academic teaching subjects (Klassen & Chiu, 2011; Klempin et al., 2019). However, as these studies included measures of *teacher* self-efficacy rather than *practice* self-efficacy for student teachers, the results are not directly comparable.

4.4. Limitations and Strengths

The study has one main limitation in the form of the cross-sectional design of the study. This means that we cannot say anything about changes and thus the direct effect of the first year of the teacher education program, but only differences for students who had just started in the first year of study and who had not been in field practice in schools as yet and another group of students one year later, i.e., students who had completed the first year and the first field placement in school. On the other hand, the design is useful for study programs if they, for example, desire to tap into students' PSE at the start of every study year to see where they can/need to introduce self-efficacy-enhancing learning experiences. Another limitation relates to the study's generalizability. Since the sample only includes students from one university college, we cannot determine how well these results reflect the wider student population across other university colleges in Denmark. For example, university colleges may differ in local teaching practices, supervision, culture or simply student background which may result in somewhat different results with regard to the differences between first- and second-year students. Thus, we would welcome future work that assesses these differences in larger samples across institutions and potentially borders.

A minor limitation is the possibility of a cohort effect, as data were collected across three years of student intake. While we find it unlikely, we acknowledge the possibility.

Future research would benefit from also including third- and fourth-year students in such cross-sectional comparisons, and within these studies also test for invariance across study years. In addition, we suggest that longitudinal studies of changes in PSE across the duration of teacher education programs are conducted in the future.

5. Conclusions

This study suggests that completion of the first year of teacher education is associated with higher practice self-efficacy on some dimensions among student teachers. In particular, results show that second-year students are more confident than first-year students in relation to teacher and teaching practices belonging to the *Planning and preparation PSE*, *Teaching in itself PSE*, *Evaluation and Differentiation PSE* dimensions, and—to a lesser extent—*Class management PSE*. Notably, the exception is *Adult collaboration PSE*, where first-year students reveal more confidence. Exploratory analyses also revealed that students who major in English were less confident in specific skills—mainly in *Planning and preparation PSE* and *Teaching in itself PSE*—whereas Mathematics majors showed more confidence in *Adult collaboration PSE*. Taken together, the findings suggest that study year is associated with practice self-efficacy in most dimensions. This indicates that teacher education affects practice self-efficacy and that students' confidence and worries vary by major.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/psycholint8020024/s1, Supplement S1: Fit of the individual items to the respective PSE subscale models. Supplement S2: Testing the necessity of local dependence and differential item functioning interaction terms in the graphical loglinear models and testing the adequacy of all final models. Supplement S3: Item category (PCM) threshold, locations, midpoints and item targets. Supplement S4: Information for conversion between sum scores and theta scores as well as information for the DIF-equation of the sum scores. Supplement S5: Test statistics from collapsibility analyses across subgroups denied by major teaching subjects.

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Institutional Review Board Statement: The study was conducted in accordance with the ethical standards of the Helsinki Declaration. All participants received written information about the project aims, including details on the voluntary nature of participation, confidentiality of their data, and compliance with GDPR for data storage and use. Ethical review and approval were waived for this study due to the non-interventional nature of the survey.

Informed Consent Statement: All participants received written information about the project aims, including details on the voluntary nature of participation, confidentiality of their data, and compliance with GDPR for data storage and use.

Data Availability Statement: The dataset presented in this article is not readily available because the data are part of an ongoing study. Requests to access the datasets should be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

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